
25 YEARS LATER: IS THE CLEAN AIR ACT OF 1999 STILL RELEVANT?

ALVIN M. TEODORO

College of Law, West Visayas State University

One of the most human and environmental health issues that the world has faced over the years is air pollution. In the Philippines, air pollution has been the top problem, especially in Metro Manila. The major contributors to the emission are motor vehicles, power plants, industrial processes, and the combustion of biomass, coal, and kerosene.

To prevent pollution, the Clean Air Act, also known as Republic Act No. 8749 (R.A. No. 8749), was signed by former president Joseph Estrada on June 23, 1999, becoming one of the major environmental laws in the Philippines.¹

With a focus on the "polluters must pay" idea, the Clean Air Act seeks to guarantee a "balanced and healthful ecology" for all Filipinos, in line with "the rhythm and harmony of nature." In carrying out the law, the Department of Environment and Natural Resources (DENR) serves as the "lead agency." Each airshed will have a common action plan and "a system of planning and coordination" to control the air quality more effectively. Pursuant to the law, the nation is divided into various airsheds based on "areas with similar climate, meteorology, and topology which affect the interchange and diffusion of pollutants in the atmosphere, or areas which share a common interest or face similar development programs, prospects, or problems."²

Legislative Framework

Key Provisions of the Act

The Clean Air Act of 1999 was legislated in order to ensure the proper implementation of government initiatives to lessen air pollution to protect public health and the environment. This act aims to reduce the harmful emissions coming from vehicles, industrial facilities, and other pollutants. The key provisions of the act focus on the Air Quality Monitoring and Information Network, National ambient air quality standards, and regulations for pollution From Stationary Sources and from Motor Vehicles.

¹ Alexa Basa, Clean Air Act after 23 years: Time to revisit the law <https://mb.com.ph/2022/11/12/clean-air-act-after-23-years-time-to-revisit-the-law>

² *Id.*

a. Air Quality Monitoring and Information Network

Section 6, Article 1 of the Clean Air Act of 1999 is the Air Quality Monitoring and Information Network which provides that the Department of Environment and Natural Resources Shall prepare an annual National Air Quality Status Report. Its purpose is to provide various measures and strategies to reduce air pollution and maintain healthy air quality levels, educate the public and provide awareness of the issues involved in maintaining and improving air quality, and have a systematic process to collect, analyze, and disseminate information about air quality levels across the country.³

The annual National Air Quality Status Report shall include, but shall not be limited to the following:

- a) Extent of pollution in the country, per type of pollutant and per type of source, based on reports of the Department's monitoring stations;
- b) Analysis and evaluation of the current state, trends and projections of air pollution at the various levels provided herein;
- c) Identification of critical areas, activities, or projects which will need closer monitoring or regulation;
- d) Recommendations for necessary executive and legislative action; and
- e) Other pertinent qualitative and quantitative information concerning the extent of air pollution and the air quality performance rating of industries in the country.⁴

Using this report, it can be ensured that the Department of Environment and Natural Resources will properly monitor data, provide accurate public information, and impose regulations that will ensure improvement for a healthier environment.⁵

b. National Ambient Air Quality Standards (NAAQS).

One of the fundamental parts of this legislation is the creation of Ambient Air Quality Guideline Values and Standards, which operate as the standard for evaluating and controlling the air quality to which the general public is exposed.

Section 12, Article 2 of the Clean Air Act of 1999 set forth that the Department of Environment and Natural Resource, in coordination with other concerned agencies, shall review and/or revise and publish annually a list of

³ Philippine Clean Air Act of 1999, Republic Act No. 8749 <https://emb.gov.ph/wp-content/uploads/2015/09/RA-8749.pdf>

⁴ *Id.*

⁵ *Id.*

hazardous air pollutants with corresponding ambient guideline values and/or standard necessary to protect public health and safety, and general welfare.⁶

These guidelines, values, and standards are designed to ensure the protection of the public against the detrimental effects of air pollution. This is based on the latest scientific knowledge and is based on the World Health Organization (WHO) standard. These include the information on:

- a) Variable factors, including atmospheric conditions, which of themselves or in combination with other factors may alter the effects on public health or welfare of such air pollutants;
- b) The other types of air pollutants that may interact with such pollutants to produce an adverse effect on public health or welfare; and
- c) The kind and extent of all identifiable effects on public health or welfare which may be expected from the presence of such pollutants in the ambient air, in varying quantities.

The National Ambient Air Quality Standards is a strong help for the Philippine government to achieve the sustainable development goals for the well-being of the Filipino people. This provision ensures that the standards set by the government are aligned with the international standard and it is carefully assessed and complied with.⁷

c. Regulation of Pollution from Stationary Sources

Section 19, Article 3 of the Clean Air Act of 1999 aims to address the air pollution coming from stationary sources such as factories, industrial facilities, and power plants. These stationary sources emit large quantities of harmful pollutants including sulfur dioxide (SO₂), nitrogen oxides (NO_x), particulate matter (PM), and volatile organic compounds (VOCs). This provision mandates the emission standards on mass rate of emission for all stationary sources of air pollution which should be based on internationally-accepted standards. The standards, whichever is applicable, shall be the limit on the acceptable level of pollutants emitted from a stationary source for the protection of the public's health and welfare. With respect to any trade, industry, process, and fuel-burning equipment or industrial plant emitting air pollutants, the concentration at the point of emission shall not exceed the limits provided in the statute.⁸

In addition, Incineration which is defined as the burning of municipal, bio-medical, and hazardous wastes, that emit poisonous and toxic fumes, is declared prohibited in Section 20 of the same act.⁹

⁶ Philippine Clean Air Act of 1999, Republic Act No. 8749, <https://emb.gov.ph/wp-content/uploads/2015/09/RA-8749.pdf>

⁷ *Id.*

⁸ *Id.*

⁹ *Id.*

d. Regulation of Pollution from Motor Vehicles

One of the major goals of the act is to mitigate air pollution from vehicular sources. Vehicles such as cars, trucks, jeepneys, and buses are emitting harmful substances such as nitrogen oxides, carbon monoxide, and particulate matter. This provision addresses air pollution concerns by providing a comprehensive framework of regulations and standards designed to reduce emissions from motor vehicles.¹⁰

Section 21, Article 4 of the Clean Air Act of 1999 provides that “for Pollution from Motor Vehicles, the DOTC shall implement the emission standards for motor vehicles. To further improve the emission standards, the DENR shall review, revise, and publish the standards every two (2) years, or as the need arises. It shall consider the maximum limits for all major pollutants to ensure substantial improvement in air quality for the health, safety and welfare of the general public. The standard for exhaust emission limit for gaseous pollutants shall be applicable for light-duty vehicles, light commercial vehicles, and heavy-duty vehicles. This section of the act also highlighted that the fuel evaporative emission for spark-ignition engines shall not exceed 2.0 grams of hydrocarbons per test.”¹¹

Furthermore, Article 4, Section 22 of this Clean Air Act prescribes for the Regulation of All Motor Vehicles and Engines. It strengthens the registration requirements of any imported new or locally-assembled new motor vehicle which shall not be registered unless it complies with the emission standards, as evidenced by a Certificate of Conformity (COC) issued by the Department.¹²

International Context

a. Comparative Analysis of the Clean Air Act Amendment of 1990 by the United States and the Clean Air Act of 1999 of the Philippines

The Clean Air Act of Amendment of the United States 1990 and the Clean Air Act of 1999 of the Philippines are environmental laws that aim to protect the welfare of its people by ensuring air quality. However, they differ in terms of scope, implementation, and key provisions.

Scope

The U.S. The Clean Air Act originally passed in 1963 and significantly amended in 1970, 1977, and 1990, addresses a wide range of air quality issues across

¹⁰ Philippine Clean Air Act of 1999, Republic Act No. 8749, <https://emb.gov.ph/wp-content/uploads/2015/09/RA-8749.pdf>

¹¹ *Id.*

¹² *Id.*

the country. It aims to protect and enhance the nation's air resources to promote public health and welfare and the productive capacity of its population.¹³

On the other hand, The Clean Air Act of the Philippines of 1999 was primarily enacted to solve the country's urgent air pollution problems, especially in major cities like Metro Manila. Its primary objective is to protect and advance the right of the people to a balanced and healthful ecology by addressing air pollution issues.

Implementation

The regulating body in The Clean Air Act of the United States is the Environmental Protection Agency (EPA) which focuses on the Extensive coverage of stationary sources, mobile sources, and hazardous air pollutants. Included in the provisions are the setting of National Ambient Air Quality Standards (NAAQS), New Source Performance Standards (NSPS), and vehicle emissions standards. The act emphasizes transparency and public access to information by making the information obtained by the Board available to the public, except information that would be harmful to a person's competitive position. In terms of public participation, the act includes mechanisms for public engagement and citizen lawsuits to enforce compliance.¹⁴

In the Philippines, the Department of Environment and Natural Resources (DENR) is the regulation body of The Clean Air Act of 1999. The law includes both stationary sources (factories, power plants) and mobile sources (motor vehicles) which focus on setting air quality standards, promoting cleaner technologies, and encouraging alternative energy sources. Public participation is encouraged by placing a strong emphasis on community involvement and transparency, with provisions for public access to air quality data and participation in environmental assessments.¹⁵

Focus Areas

The Philippine Clean Air Act of 1999 focuses on highlighting the connection between sustainable development and air quality objectives. It also aims to lessen reliance on fossil fuels by supporting alternative energy sources. And it has a significant focus on community and public involvement in air quality management.¹⁶

¹³ Evolution of the Clean Air Act, [from https://www.epa.gov/clean-air-act-overview/evolution-clean-air-act](https://www.epa.gov/clean-air-act-overview/evolution-clean-air-act)

¹⁴ United States H.R.3030 - Clean Air Act Amendments of 1990.

¹⁵ Philippine Clean Air Act of 1999, Republic Act No. 8749, <https://emb.gov.ph/wp-content/uploads/2015/09/RA-8749.pdf>

¹⁶ *Id.*

In comparison to the Clean Air Act of 1990 by the United States, it focuses on a comprehensive approach covering a wide range of pollutants and sources. Also, it uses a combination of regulatory measures and market-based approaches to achieve air quality goals and emphasizes technological innovation and continuous improvement in pollution control measures.¹⁷

In a nutshell, both the Clean Air Act of the Philippines and the Clean Air Act of the United States aim to safeguard public health and ensure quality air across the nation. The Philippine law is primarily concerned with incorporating sustainable development and public engagement while the U.S. law is broader in scope, with a longer history of amendments and a more detailed regulatory framework that includes market-based mechanisms.

Alignment to the United Nations Sustainable Development Goals (UNSDG)

The 2030 Agenda for Sustainable Development, adopted by all United Nations Member States in 2015, provides a shared goal of achieving peace and prosperity for the people and planet. At its heart are the seventeen (17) Sustainable Development Goals (SDGs), which are an urgent call for action by all countries - developed and developing - in a global partnership. One of its main goals is to attack climate change and preserve the environment.¹⁸

The R.A. No. 8749 “Clean Air Act of 1999” aligns with several goals of the United Nations Sustainable Development Goals (UNSDGs), reflecting its comprehensive approach to improving air quality, protecting public health, and promoting sustainable development.

The Clean Air Act of 1999 has a significant positive impact on several goals. It is directly contributing to SDG “Good Health and Well-Being” by decreasing the incidence of respiratory and cardiovascular diseases associated with poor air quality. Furthermore, by regulating the emission standards and promoting cleaner technologies the health risks can be reduced and it can help in improving the well-being of the people. SDG 7 which is “Affordable and Clean Energy” can also be achieved by promoting the use of cleaner fuels and the development of renewable energy sources.¹⁹

¹⁷ United States H.R.3030 - Clean Air Act Amendments of 1990, <https://www.congress.gov/bill/101st-congress/house-bill/3030#:~:text=Prohibits%20any%20lead%20content%20and,when%20using%201990%20certification%20fuel>

¹⁸ United Nations Sustainable Development, <https://sdgs.un.org/goals>

¹⁹ Sustainable Development Goals and air pollution, <https://www.cleanairfund.org/news-item/sustainable-development-goals/>

In addition, the act is indirectly contributing to SDG 11, “Sustainable Cities and Communities” by regulating the emissions from industrial activities and vehicles. It can also help in achieving SDG 12 or “Responsible Consumption and Production” because the act mandates the use of pollution control technologies and cleaner production processes, aligning with the goals of minimizing waste and promoting sustainable consumption and production. By reducing emissions and encouraging responsible environmental practices, the act supports the broader agenda of sustainable management of resources and waste. Lastly, SDG 13 “Climate Action” affected the reduction of emissions of greenhouse gasses and other pollutants. Hence, mitigating the risks of climate change.²⁰

The Clean Air Act of 1999 is undeniably aligned with the United Nations Development Goals as it addresses air pollution, promotes public health, encourages the use of clean energy, supports sustainable urban development, fosters responsible consumption and production, combats climate change, and protects terrestrial ecosystems. With the proper implementation program and stringent application of its regulations, cleaner air can be achieved.

Stakeholders involvement and initiatives

Acknowledging the importance of the Clean Air Act of 1999 and the rigid implementation of the programs of the Department of Environment and Natural Resources, the stakeholders are becoming more involved and conducting initiatives to help address the issues on air pollution.

The National Air Quality Status Report 2016-2018 of the Department of Environment and Natural Resources (DENR) showed that the regional agencies are becoming more involved in air quality management, as reflected in the completion of emissions inventories and regional monitoring results.²¹

There are also initiatives from other government agencies such as the Department of Energy (DoE) and the Department on Trade and Industries (DTI) on the standards for fuels and additives; the Department of Transportation (DoTr) on their programs for Euro 4 standards for fuels and engines; and Department of Health on the implementing guidelines of Inter-agency Committee of Environmental Health.²²

For the health aspect, the Lung Center of the Philippines and the World Health Organization (WHO) initiated a project on the Air Pollution Monitoring

²⁰ Sustainable Development Goals and air pollution, <https://www.cleanairfund.org/news-item/sustainable-development-goals/>

²¹ National Air Quality Status Report 2016-2018, <https://air.emb.gov.ph/wp-content/uploads/2021/07/NAQSR-2016-2018-FV.pdf>

²² *Id.*

Project which provides assistance on the PM_{2.5} breakpoints. In the academe, the University of the Philippines Diliman, De La Salle State University, and local and international academies have contributed to aerosol characterization research. For non-governmental organizations, there are several projects which assist in measurement and action plans.²³

Limitations and Challenges

The Clean Air Act of 1999 has significant actions in addressing air pollution in the country, however, it cannot be denied that there are several limitations and challenges that affect the effective implementation of the act.

The National Air Quality Status Report 2016-2018 of the Department of Environment and Natural Resources (DENR) shows that despite the consolidated efforts to improve air quality status, there are still remaining challenges and needed actions in air quality management.

Some of the notable challenges based on the DNRs report are the following:

- Lack of Air Quality Index (AQI) Breakpoints for PM_{2.5}
- Lack of siting criteria for air quality monitoring stations
- Lack of Data Handling Guidelines for Ambient Air Quality Monitoring Data
- Dependence expensive reference/equivalent AQM equipment
- Lack of an On-Line Real-Time Data Acquisition System in monitoring Large Industrial Facilities with Continuous Emission/ Opacity Monitoring Systems (CEMS/COMS)
- Lack of Online System for the Issuance of Permit to Operate (PTO) for Air Pollution Source Installation (APSI)
- Decreasing the uncertainties in emissions inventory
- Unable to capture the 75% data capture rate
- Assessment of air quality and its health impact at the regional level
- Weak inter-agency participation on air quality and health actions
- Management of mobile emissions sources²⁴

These limitations suggest that the actual and effective implementation of the Clean Air Act of 1999 after how many years is still a challenge. These challenges are mostly caused by weak monitoring systems; these are gaps in the air quality monitoring network, including inadequate coverage and outdated equipment, which hinder accurate and comprehensive assessment of air quality across the country. There are also coordination issues among the government agencies and the LGUS leading to fragmented efforts and overlapping responsibilities. The economic and technological barriers are also the contributing factors. The high cost for compliance triggers the non-compliance of business from different industries, particularly small and medium enterprises (SMEs) due to lack of financial resources to invest in the

²³ National Air Quality Status Report 2016-2018, <https://air.emb.gov.ph/wp-content/uploads/2021/07/NAQSR-2016-2018-FV.pdf>

²⁴ *Id.*

required upgrades. The technological limitations of the Department of Environment and Natural Resources is also one of the barriers because the access to and adoption of advanced pollution control technologies are limited, particularly in rural and economically disadvantaged areas.

Conclusion

The Philippine Clean Air Act of 1999, Republic Act No. 8749 remains to be relevant for more than two decades after its enactment. Its comprehensive framework which aims to improve air quality in the country is essential in the protection of public health. Its alignment with the United Nations Sustainable Development Goals (UNSDGs) provides more importance that this act must be upheld to further improve the air quality in the Philippines. The Clean Air Act of 1999, even if it is timely and relevant, still faces a lot of limitations and challenges. Even though the DENR provided strategic actions and timelines to combat these predicaments, it cannot be denied that these occurrences happened ever since the effectivity of the Philippine Clean Air Act of 1999. These challenges will not be addressed unless the government strengthens the enforcement and compliance of the law, increases in the resources for regulatory bodies, and investments to advance technology in such a way as to ensure comprehensive and accurate data collection.

In the end, it is the government and its people that is responsible for taking good care of the environment and ensuring clean air quality. The Republic Act No. 8749 or the Philippine Clean Air Act of 1999 can be effectively and efficiently implemented when the government agencies and stakeholders collaborate to achieve the law's goal of cleaner air and a more environmentally friendly atmosphere for the common good and public welfare.

Bibliography

Statutes

An Act Providing for A Comprehensive Air Pollution Control Policy and For Other Purposes, Republic Act No. 8749,
<https://emb.gov.ph/wp-content/uploads/2015/09/RA-8749.pdf>

United States H.R.3030 - Clean Air Act Amendments of 1990,
<https://www.congress.gov/bill/101st-congress/house-bill/3030#:~:text=Prohibits%20any%20lead%20content%20and,when%20using%201990%20certification%20fuel.>

Reports

National Air Quality Status Report 2016-2018, Department of Environment and Natural Resources,
<https://air.emb.gov.ph/wp-content/uploads/2021/07/NAQSR-2016-2018-FV.pdf>

Online Sources:

Clean Air Act after 23 years: Time to revisit the law,
<https://mb.com.ph/2022/11/12/clean-air-act-after-23-years-time-to-revisit-the-law>

Evolution of the Clean Air Act <https://www.epa.gov/clean-air-act-overview/evolution-clean-air-act>

United Nations Sustainable Development,
<https://sdgs.un.org/goals>

Sustainable and Developmental Goals,
<https://www.cleanairfund.org/news-item/sustainable-development-goals/>